

Physics 180Q Final Project: Demonstration of the Quantum Zeno Effect

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This is my wonderful abstract.

Introduction—

*Theory—*The quantum Zeno effect will be demonstrated here by considering the evolution between polarization states. The basis states in this case are $|H\rangle$ and $|V\rangle$. An arbitrary linear polarization state, then, can be written as

$$|\psi\rangle = \cos(\theta) |H\rangle + \sin \theta |V\rangle, \quad (1)$$

where θ is the angle of polarization.

We then aim to calculate the magnitude of the state that transmits through the polarizer. Allowing the photon to interact with the polarizer is effectively “measuring” the state of the photon, which causes its wavefunction to collapse into $|H\rangle$. Thus, in other words, we want to calculate the magnitude of state projected onto the polarizer axis. Without loss of generality, we will assume the polarizer is set to the $|H\rangle$ axis.¹ The projection operator onto the $|H\rangle$ state is

$$P_H = |H\rangle \langle H|. \quad (2)$$

Thus, projecting the arbitrary linear polarization onto $|H\rangle$ gives

$$\begin{aligned} P_H |\psi\rangle &= |H\rangle \langle H| (\cos(\theta) |H\rangle + \sin \theta |V\rangle) \\ &= \cos(\theta) |H\rangle. \end{aligned} \quad (3)$$

The magnitude of this state is

$$\langle \psi | \psi \rangle = \cos^2(\theta), \quad (4)$$

which gives the probability that a given photon is transmitted. If there is a stream of photons entering a detector,

$$\langle \psi | \psi \rangle \propto \Gamma, \quad (5)$$

where Γ is the coincidence count rate. Thus, the relation between the input and output count rate is given by

$$\Gamma(\theta) = \Gamma_{\max} \cos^2(\theta). \quad (6)$$

For classical theories of light as an electromagnetic wave, this relation considering $\Gamma \leftrightarrow I$ is also known as Malus’s law.

In this experiment, we wish to verify this relation is true for single photons as it traverses multiple polarizers, each collapsing the photon’s wavefunction into its polarization direction. If there are n polarizers set so that the angle θ between them is consistent, this is equivalent to projecting successively onto different polarization states. Thus, the coincidence count rate for n polarizers with angle difference θ is

$$\Gamma(\theta, n) = \Gamma_{\max} \cos^{2n}(\theta). \quad (7)$$

In the case where the total angle difference is 90° , the relation between n and θ is

$$\theta = \frac{90^\circ}{n} = \frac{\pi}{2n}. \quad (8)$$

Thus, the final expression for the coincidence count rate for the experiment is expected to be

$$\Gamma(n) = \Gamma_{\max} \cos^{2n} \left(\frac{\pi}{2n} \right), \quad (9)$$

or equivalently,

$$\Gamma(\theta) = \Gamma_{\max} \cos^{\frac{\pi}{\theta}}(\theta). \quad (10)$$

It is critical to note that for increasing n , the rate of coincident photons also increases. Interestingly, for $n \rightarrow \infty$, $\Gamma \rightarrow \Gamma_{\max}$, meaning that all photons pass through despite the initial and final polarization angles being 90° apart. Thus, the quantum Zeno effect predicts that as more polarizers are added, the more frequent “observations” prevent the quantum state from diverging from its observed state. In determining evidence for the quantum Zeno effect, this model will be compared to the model where the effect does not exist, or a flat line.

Experimental Methods—

Conclusion—

Summary—

¹ If the polarizer set at a different angle α , this is equivalent to rotating the polarization state by $-\alpha$.